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Electron Paramagnetic Resonance in Mixed Crystals $(BaF_2)_{1-x}(LaF_3)_x$ Activated by Ce^{3+} Ions

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Abstract—The electron paramagnetic resonance (EPR) spectra of mixed crystals $(BaF_2)_{1-x-y}(LaF_3)_x(CeF_3)_y$ ($y = 0.001 = 0.1\%$, $x = 0$–0.02) are investigated in a magnetic field $\mathbf{H} \parallel \mathbf{C}_4$ at a frequency of 9.5 GHz. The angular dependence of the EPR spectrum is measured for the sample with $x = 0.02$. The lines attributed to Ce^{3+} impurity centers with tetragonal symmetry and g factors ($g_\parallel = 0.75$, $g_\perp = 2.4$) close to those measured for the $KY_3F_{10} : Ce^{3+}$ compound are separated in the complex EPR spectrum. The assumption is made that the aforementioned impurity centers are cubooctahedral clusters of the La_6F_{37} type in which one of the La^{3+} ions is replaced by the Ce^{3+} ion.
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1. INTRODUCTION

Direct observations of the formation of rare-earth (*RE*) clusters in paramagnetic crystals $(MeF_2)_{1-x}(REF_3)_x$ with the use of electron paramagnetic resonance (EPR) have been practically impossible because of the considerable broadening of the resonance lines at a trifluoride content above $x \approx 0.1\%$. From this standpoint, binary solid solutions $(MeF_2)_{1-x-y}(RF_3)_x(REF_3)_y$ with diamagnetic ions *R* (*R* = La, Lu, Y) at a low content *y* of the paramagnetic rare-earth component seem to be more convenient objects. Kazanskiĭ [1] analyzed optically detected EPR in absorption bands for a series of fluoride compounds and made the inference that the cluster formation is determined by the total trifluoride content $x + y$ and that the EPR spectra of RE^{3+} ions in these compounds at a total content $x + y \geq 0.001$ are associated with the tetragonal positions of diamagnetic ions in the R_6F_{37}-type clusters. In order to clarify the potentials of the EPR method in more detail, it is expedient to analyze the EPR spectra of mixed crystals $(MeF_2)_{1-x-y}(RF_3)_x(REF_3)_y$ with trifluoride contents *x* and *y* varying over the widest possible ranges.

In our recent work [2], we studied the EPR spectra of mixed crystals $(BaF_2)_{1-x}(LaF_3)_x$ activated with 0.1% Yb^{3+}. It was found that, even at a low content $x = 0.001$, the EPR spectra exhibit additional resonance lines due to the formation of trigonal clusters involving two trivalent ions in barium positions and two compensating fluorine ions of the La^{3+}–F^-–Yb^{3+}–F^- type. Analysis of these lines was complicated by the fact that the initial EPR spectrum, which was observed even at $x = 0$, had a hyperfine structure caused by ^{171}Yb and ^{173}Yb isotopes. However, we did not reveal indications of the lines associated with tetragonal impurity centers that could be identified with Yb^{3+} ions in the La_6F_{37} clusters.

In the present work, we studied the same mixed crystals $(BaF_2)_{1-x}(LaF_3)_x$ but with Ce^{3+} ions serving as paramagnetic probes. The Ce^{3+} ions do not have magnetic isotopes; consequently, the EPR spectrum of Ce^{3+} does not exhibit a hyperfine structure.

2. SAMPLES AND EXPERIMENTAL TECHNIQUE

In our experiments, we used 0.1% Ce^{3+}-activated samples at trifluoride contents $x = 0$, 0.001, 0.002, 0.005, 0.01, and 0.02. The EPR spectra were measured by the stationary method on an IRÉS-1003 spectrometer at a frequency of ≈9.5 GHz and at a temperature of ≈15 K. In addition, the samples at trifluoride contents $x = 0$ and 0.02 were examined using the pulsed method of electron spin echo at a frequency of ≈9.8 GHz and at a temperature of ≈7 K.

3. RESULTS AND DISCUSSION

For pure BaF_2 ($x = 0$), we revealed a single EPR center with tetragonal symmetry ($g_\parallel = 2.601$, $g_\perp = 1.555$; for comparison, see the results obtained earlier in [3]). In a magnetic field applied parallel to the fourfold symmetry axis ($\mathbf{H} \parallel \mathbf{C}_4$), the characteristic EPR spectrum exhibits a single line at a magnetic field strength of ≈2.603 kG and a double line at a magnetic field strength

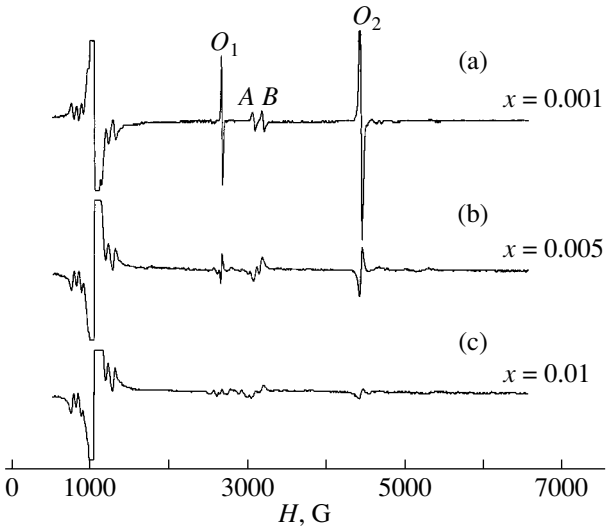

Fig. 1. Ce^{3+} EPR spectrum of mixed crystals $Ba_{1-x}La_xF_{2+x}$ + 0.1%Ce^{3+} at different contents x in the magnetic field **H** || $\mathbf{C}_4$. Lines O_1 and O_2 correspond to the Ce^{3+} tetragonal impurity center. Lines *A* and *B* are the most intense additional EPR lines at low contents x. The intense line observed in a magnetic field of ~1 kG is the EPR spectrum of the reference sample CaF_2 + 0.8% Er^{3+}.

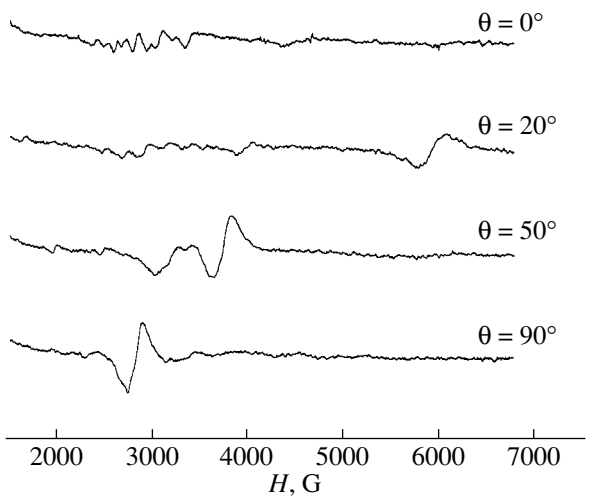

Fig. 2. EPR spectrum of the $Ba_{1-x}La_xF_{2+x}$ + 0.1%Ce^{3+} (x = 0.02) sample at different angles θ in the plane C_4–C_3–C_2. The angle θ = 0 corresponds to the magnetic field **H** || $\mathbf{C}_4$.

of 4.355 kG (lines O_1 and O_2 in Fig. 1, respectively). It is worth noting that the intensity of the EPR line associated with Ce^{3+} ions in pure BaF_2 (x = 0) is several times lower than that of Yb^{3+} ions in pure BaF_2. At x = 0.001, the intensity of the line attributed to the initial Ce^{3+} tetragonal center decreases. Moreover, there appear additional lines, among which the two most intense lines are observed at magnetic field strengths H = 3.070 and 3.188 kG (lines *A* and *B* in Fig. 1a, respectively). A further increase in the LaF_3 content leads to a substantial decrease in the intensity of the initial lines attributed to the tetragonal impurity centers (by approximately one order of magnitude at x = 0.01; see Fig. 1c) and to the appearance of new additional lines at magnetic field strengths in the range 2.5–3.2 kG. In order to determine the actual concentration dependence of the intensity of the EPR lines of interest, we measured the EPR spectra of $Ba_{1-x}La_xF_{2+x}$ samples at different contents x and simultaneously recorded the EPR spectrum of the reference sample. As a reference sample, we used a CaF_2 + 0.8% Er single crystal. This crystal is a convenient reference sample, because its EPR spectrum is characterized by only one intense line assigned to the Er^{3+} cubic impurity center (i.e., the isotropic EPR line with g = 6.78, H ~ 1.0 kG, and a linewidth of ≈40 G, which is close to the linewidth for the Ce^{3+} ions under investigation).

As was noted above, the sample at a trifluoride content x = 0.02 was studied in more detail. The electron spin echo measurements demonstrated that the spectrum of this sample does not contain the broad isotropic line, which was previously observed in the spectra of the $(BaF_2)_{1-x}(LaF_3)_x$ + 0.1% Yb^{3+} ($x \geq 0.01$) samples [2]. Thus, there are grounds to believe that no paramagnetic centers whose environment has a disordered structure are formed in the $(BaF_2)_{1-x}(LaF_3)_x$ + 0.1% Ce^{3+} ($x \approx 0.02$) samples studied in the present work.

Analysis of the angular dependence of the EPR spectrum for the sample at x = 0.02 (Fig. 2) revealed additional resonance lines that correspond to paramagnetic centers of tetragonal symmetry with $g_\parallel \approx 0.75$ and $g_\perp$ = 2.4. The width of the lines attributed to these centers (~200 G) is one order of magnitude larger than that of the *O* centers, whereas the intensity of these lines is comparable to the intensity of the lines assigned to the *O* centers in the sample at x = 0. The former centers can be associated with the formation of La_6F_{37} clusters. This is confirmed by the fact that the g factors obtained in our case are close to those previously observed for the KY_3F_{10} : Ce^{3+} compound, whose lattice involves Y_6F_{36} cubooctahedral clusters with Ce^{3+} ions occupying Y^{3+} sites ($g_\parallel$ = 0.77, $g_\perp$ = 2.46 [4]).

4. CONCLUSIONS

Thus, the EPR spectra of mixed crystals $(BaF_2)_{1-x}(LaF_3)_x$ indicate that the structure of these crystals can involve La_6F_{37} cubooctahedral clusters. However, the appearance of additional EPR lines is not necessarily associated with the formation of these clusters in the crystal structure. A more complete identification of the EPR spectra calls for further investigation.

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